An Overview of Openstack Architecture

Tiago Rosado
Polytechnic Institute of Coimbra
ISEC – Coimbra Institute of Engineering
Rua Pedro Nunes, 3030-199 Coimbra, Portugal
Tel. ++351 239 790 200
a20105059@alunos.isec.pt

Jorge Bernardino
Polytechnic Institute of Coimbra
ISEC – Coimbra Institute of Engineering
Rua Pedro Nunes, 3030-199 Coimbra, Portugal
Tel. ++351 239 790 200
jorge@isec.pt

ABSTRACT
Cloud Computing concept refers to both the applications delivered as services over the Internet and the servers and system software in the datacenters that provide those services. These solutions offer pools of virtualized computing resources, paid on a pay-per-use basis, and drastically reduce the initial investment and maintenance costs. Efficient and flexible resource management is the main focus for the cloud solutions on the market as well as scalability and adaptability to new environments. Openstack exceeded the market as a scalable, performant and highly adaptive open source architecture for both public and private cloud solutions as well as leveraging from hardware resources either they be professional or entry level. This paper gives an overview of Openstack software components functionalities in order to design and implement unique cloud computing solutions to fit enterprises purposes.

Categories and Subject Descriptors
C.5 [**Computer System Implementation**]: Miscellaneous
D.2.11 [**Software Engineering**]: Software Architectures – Domain-specific architectures

General Terms
Design, Experimentation, Performance.

Keywords
Openstack, IaaS, Cloud Computing, Open Source.

1. INTRODUCTION
Cloud computing have gained more and more attention in recent years, and open source solutions in general have been growing in terms of quality, security and internal requirements. Leading the trend on open source in the IT scenarios are the cloud computing platforms. Cloud computing is a model for delivering ubiquitous, on-demand computer resources over the Internet, relying on essential characteristics as on-demand self-service, broad network access, resource pooling, elasticity, measuring and monitoring. All resources like storage, computing or network are presented to final users as services in the form of infrastructure, platform or software. This emerging model can be applied inside enterprises, known as private clouds, or acquired as services form external vendors, called public clouds.

IDEAS '14, July 07 - 09 2014, Porto, Portugal

http://dx.doi.org/10.1145/2628194.2628195

OpenStack is the open source cloud computing platform most widely adopted in industry. Meant to be simple to implement, this Infrastructure as a Service (IaaS) is open and massively scalable. There are many studies focusing on open source solutions for cloud computing, presenting detailed comparisons and overviews between them [5] [1]. Others exist pointing studies on modules in particular [6] or detailed success implementations of Openstack on education and industry [2][3]. This paper gives an update, highlight and detailed overview of Openstack architecture, showing the essential services that are necessary to install.

2. OPENSTACK ARCHITECTURE
Openstack is a fully open sourced combination of software projects developed originally by NASA and Rackspace, having gained so far such reliability robustness and availability as an IaaS solution for both public and private cloud infrastructures [4]. Its modular and highly configurable architecture enables this solution to be conceived and tailored to fit available hardware resources either they be professional or entry level, with few or more computer nodes, in response to each case in particular. This allows enterprises to choose from a variety of complementary services in order to meet different needs regarding to computing, networking and storage [7]. Figure 1 represents the Openstack conceptual architecture with all native software components, developed by companies and individual supporters, depicting how they interact with each other.

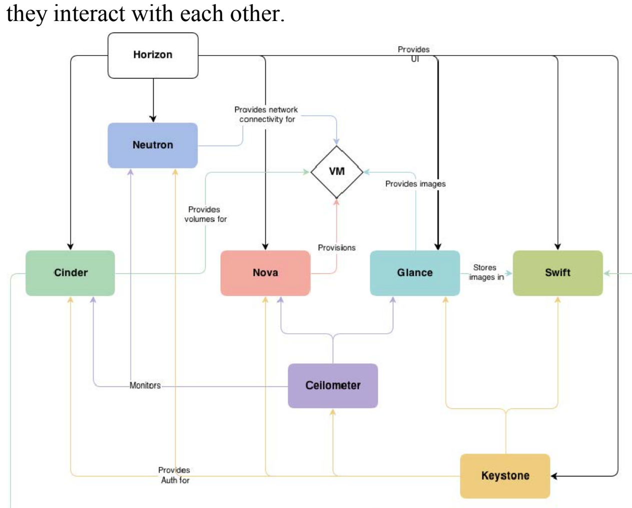

Figure 1 - Openstack conceptual architecture

Following we describe those components dividing them into five groups due to their characteristics such as computing, networking, storing, shared and supporting services. Although the supporting services are not native from Openstack's core, are essential for its operation [8]. Services can be installed in accordance with requirements, which mean that we can install all or only a few.

2.1 Computing

Nova: Also known as Openstack Compute or Nova Compute, it provides virtual servers upon demand interacting with the hypervisors, supporting several kinds such as KVM, Xen, VMware or Hyper-V. It's an array of software that provides services for cloud resource management through its APIs, capable of orchestrating running instances, networks and access control. It's an essential service for a basic cloud architecture implementation.

Glance: It's the Openstack Image service, a lookup and retrieval system for virtual machine (VM) images. It provides services for discovering, registering and retrieving virtual images through an API that allows querying of VM image metadata, catalog and manage large libraries of server images. It's an essential service for a basic cloud architecture implementation.

2.2 Networking

Neutron: The Openstack networking services is a project to provide network connectivity as a service between interface devices managed by other Openstack services (e.g. Nova). It provides capabilities for managing dynamic host configuration protocol (DHCP), static internet protocols (IP's), or virtual area networks (VLAN's) along with other advanced policies and topologies. Due to its architecture it allows users to take leverage of frameworks (e.g. intrusion detection systems, load balancing, virtual private networks) from supported vendors.

2.3 Storing

Swift: Also known as Openstack Object Storage, Swift is a highly available, distributed object/blob store. Allows users to store or retrieve files. It can be used by Cinder component to back up the VMs volumes.

Cinder: Also known as Openstack Block Storage or Cinder Volumes it provides persistent block storage (or volumes) to guest virtual machines. By working with Swift, Cinder can use it to back up the VMs volumes. In earlier releases of Openstack this service was an integral part of Nova in the form of nova-volume. Cinder will mainly interact with Nova, providing volumes for its instances, allowing through its API the manipulation of volumes, volume types and volume snapshots.

2.4 Shared Services

Keystone: The Openstack identity is a single point of integration for Openstack policy, catalog, token and authentication, applying them to users and services interactions. Without Keystone there's no compliance between services consequently the cloud solutions won't be available for end users. It's an essential service for a basic cloud architecture implementation.

Horizon: The Openstack dashboard is a modular web application that provides a user interface for cloud infrastructure management by interacting with all other services public APIs. It's an essential service for a basic cloud architecture implementation.

Ceilometer: Is a new component of the Openstack services that provides a configurable collection of metering data in terms of CPU and network costs available from all other services in the platform, delivering a unique point of contact for billing systems.

2.5 Supporting Services

Database: By default Openstack uses MySQL as its relational database management system for core services to store configurations and management information. Typically MySQL database is installed on the main node of the infrastructure, the controller node. It's an essential service for a basic cloud architecture implementation.

Advanced Message Queue Protocol: AMQP is the messaging technology used in Openstack for inter-process communication. It uses by default RabbitMQ and supports others such as Qpid or ZeroMQ. The broker used gives a common platform for processes to send and receive messages between them making use of little or no knowledge of each other's component definitions. It's an essential service for a basic cloud architecture implementation.

3. CONCLUSIONS AND FUTURE WORK

Cloud architectures exploit virtualization techniques to provision multiple Virtual Machines (VMs) on the same physical host, so as to efficiently use available resources. The cloud computing paradigm and available solutions in the market, regarding to Openstack, have been implemented and explored in various case scenarios, with major tests given in academy and industry, presenting good results for performance, reliability and final user acceptance. Openstack leverages due to its modular architecture, simplicity to implement and adaptation not only to major data centers but also to entry level hardware providing enterprises a massively scalable cloud infrastructure. Allied to the fact of being open source, which helps on cost control, enables to deliver rapidly on-demand resources to final users. This work has presented an overview of Openstack architecture, services available and how they work and adapt into any hardware infrastructure providing a sustainable and robust private cloud solution. Based on this study, future work will involve an investigation on how to improve final user experience with the platform and administration activities regarding to measures and monitoring the cloud environment.

4. REFERENCES

[1] Bist, M. Wariya, M. Agarwal, A. 2012. Comparing Delta, Open Stack and Xen Cloud Platforms: A Survey on Open Source IaaS. In 2013 3rd IEEE International Advance Computing Conference (IACC).

[2] Bonner, S. Pulley, C. Kureshy, I. Holmes, V. Brennan, J. James, Y. 2013. Using OpenStack to Improve Student Experience in an H.E. Environment. In Science and Information Conference 2013, London, UK, October 2013.

[3] Campos, I. Fernández-del-Castillo, E. Heinemeyer, S. Lopez-Garcia, A. Pahlen, F.v.d. 2013. Borges, G. Phenomenology tools on cloud infrastructures using OpenStack. The European Physical Journal C.

[4] Castillo, J.Á.L.d. Mallichan, K. Al-Hazmi, Y. OpenStack Federation in Experimentation Multi-cloud Testbeds. 2013. In 2013 IEEE ICCCTS.

[5] Laszewski, G.v. Diaz, J. Wang, F. Fox, G. C. 2012. Comparison of Multiple Cloud Frameworks. Pervasive Technology Institute, Indiana University. Bloomington, IN 47408, U.S.A.

[6] Litvinski, O. Gherbi, A. Experimental Evaluation of Openstack Compute Scheduler. 2013. In The 4th International Conference on Ambient Systems, Networks and Technologies.

[7] Openstack – [online] http://www.openstack.org/ (Accessed April 2014).

[8] Pepple, K. 2013. *Deploying Openstack, Creating open source clouds, 2nd Edition*, O'Reilly Media.